

Your Name:

ID #:

Worksheet: Limits to Infinity-Part 1

Note: for all the problems, identify the limit as a number, $-\infty$, ∞ or DNE (does not exist)

1. Please keep in mind the list of functions in order of their rate of growth – quickest to slowest:

Factorial($x!$), **Exponential** ($4^x, e^x$), **Algebraic** ($x^4, x^{0.5}$), **Logarithm** ($\log_2 x, \ln x$)

Then find the following limits:

(a) $\lim_{x \rightarrow \infty} \frac{x^3}{x!} =$

Solution:

As $x \rightarrow \infty$, $x!$ grows faster than x^3 ,

$$\lim_{x \rightarrow \infty} \frac{x^3}{x!} = 0.$$

(b) $\lim_{x \rightarrow \infty} \frac{\sqrt[3]{x^4 + 2x^3 + 1}}{\ln x + 3} =$

Solution:

Extracting dominant terms from the numerator and denominator,

$$\lim_{x \rightarrow \infty} \frac{\sqrt[3]{x^4 + 2x^3 + 1}}{\ln x + 3} = \lim_{x \rightarrow \infty} \frac{x^{\frac{4}{3}}}{\ln x}.$$

As $x \rightarrow \infty$, $x^{\frac{4}{3}}$ grows faster than $\ln(x)$, then $\lim_{x \rightarrow \infty} \frac{x^{\frac{4}{3}}}{\ln x} = \infty$.

$$\text{Hence, } \lim_{x \rightarrow \infty} \frac{\sqrt[3]{x^4 + 2x^3 + 1}}{\ln x + 3} = \infty.$$

(c) $\lim_{x \rightarrow \infty} \frac{x^{10} + 4x^7 + 10}{32x^{12} + 10} =$

Solution:

Extracting dominant terms from the numerator and denominator,

$$\lim_{x \rightarrow \infty} \frac{x^{10} + 4x^7 + 10}{32x^{12} + 10} = \lim_{x \rightarrow \infty} \frac{x^{10}}{32x^{12}} = \lim_{x \rightarrow \infty} \frac{1}{32x^2} = 0.$$

(d) $\lim_{x \rightarrow \infty} \frac{5x + 1}{x^2 - 3x + 4}$

Solution:

Extracting dominant terms from the numerator and denominator,

$$\lim_{x \rightarrow \infty} \frac{5x + 1}{x^2 - 3x + 4} = \lim_{x \rightarrow \infty} \frac{5x}{x^2} = 0.$$

(e) $\lim_{x \rightarrow -\infty} \frac{2^x}{x^2}$

Solution:

$$\lim_{x \rightarrow -\infty} 2^x = \lim_{x \rightarrow \infty} \frac{1}{2^x} = 0 \quad \text{and} \quad \lim_{x \rightarrow -\infty} x^2 = \infty$$

$$\text{Hence, } \lim_{x \rightarrow -\infty} \frac{2^x}{x^2} = \frac{\lim_{x \rightarrow -\infty} 2^x}{\lim_{x \rightarrow -\infty} x^2} = 0$$

(f) $\lim_{x \rightarrow \infty} \frac{2^x}{x^2}$

Solution:

As $x \rightarrow \infty$, 2^x grows faster than x^2 ,

$$\lim_{x \rightarrow \infty} \frac{2^x}{x^2} = \infty$$

(g) $\lim_{n \rightarrow \infty} \frac{5^n + 2}{n^9 + 100} =$

Solution:

Extracting dominant terms from the numerator and denominator,

$$\lim_{n \rightarrow \infty} \frac{5^n + 2}{n^9 + 100} = \lim_{n \rightarrow \infty} \frac{5^n}{n^9},$$

As $n \rightarrow \infty$, 5^n grows faster than n^9 , we have $\lim_{n \rightarrow \infty} \frac{5^n}{n^9} = \infty$.

$$\text{Hence, } \lim_{n \rightarrow \infty} \frac{5^n + 2}{n^9 + 100} = \infty$$

$$(h) \lim_{n \rightarrow \infty} \frac{n^4 + 10l n n}{4\sqrt{n^8 + 3n + 100}} =$$

Solution:

Extracting dominant terms from both the numerator and denominator,

$$\begin{aligned} \lim_{n \rightarrow \infty} \frac{n^4 + 10l n n}{4\sqrt{n^8 + 3n + 100}} &= \lim_{n \rightarrow \infty} \frac{n^4}{4n^{(8/2)}} \\ &= \lim_{n \rightarrow \infty} \frac{1}{4} = \frac{1}{4} \end{aligned}$$

$$(i) \lim_{n \rightarrow \infty} \frac{-n^4 + 3n^2}{19n^3 + 3n^2 - 10} =$$

Solution:

Extracting dominant terms from the numerator and denominator,

$$\lim_{n \rightarrow \infty} \frac{-n^4 + 3n^2}{19n^3 + 3n^2 - 10} = \lim_{n \rightarrow \infty} \frac{-n^4}{19n^3} = \lim_{n \rightarrow \infty} \frac{-n}{19} = -\infty$$

$$(j) \lim_{x \rightarrow -\infty} \frac{x^{120} + 20}{-x^{100} + e^x} =$$

Solution:

$$\lim_{x \rightarrow -\infty} e^x = \lim_{x \rightarrow \infty} \frac{1}{e^x} = 0$$

$$\Rightarrow \lim_{x \rightarrow -\infty} \frac{x^{120} + 20}{-x^{100} + e^x} = \lim_{x \rightarrow -\infty} \frac{x^{120} + 20}{-x^{100}}.$$

$$\lim_{x \rightarrow -\infty} \frac{x^{120} + 20}{-x^{100}} = \lim_{x \rightarrow -\infty} \frac{x^{120}}{-x^{100}} + \lim_{x \rightarrow -\infty} \frac{20}{-x^{100}}$$

$$= \lim_{x \rightarrow -\infty} -x^{20} + \lim_{x \rightarrow -\infty} \frac{20}{-x^{100}} = -\infty + 0 = -\infty$$

$$\text{hence, } \lim_{x \rightarrow -\infty} \frac{x^{120} + 20}{-x^{100} + e^x} = -\infty$$

$$\textbf{Note: } \lim_{x \rightarrow \infty} \frac{x^{120} + 20}{-x^{100} + e^x} = 0$$

$$(k) \lim_{n \rightarrow -\infty} \frac{5^n + n^2}{-e^n + 1} =$$

Solution:

$$\begin{aligned} \lim_{n \rightarrow -\infty} (5^n) &= \lim_{n \rightarrow \infty} \frac{1}{5^n} = 0, \\ \text{similarly, } \lim_{n \rightarrow -\infty} (-e^n) &= \lim_{n \rightarrow \infty} -\frac{1}{e^n} = 0, \\ \Rightarrow \lim_{n \rightarrow -\infty} \frac{5^n + n^2}{-e^n + 1} &= \frac{\lim_{n \rightarrow -\infty} (5^n + n^2)}{\lim_{n \rightarrow -\infty} (-e^n + 1)} \\ &= \frac{\lim_{n \rightarrow -\infty} (n^2)}{\lim_{n \rightarrow -\infty} (1)} = \infty \end{aligned}$$

Note: $\lim_{n \rightarrow \infty} \frac{5^n + n^2}{-e^n + 1} = -\infty$

$$(l) \lim_{x \rightarrow \infty} \frac{x^8 + \ln x}{3x^8 + 2} =$$

Solution:

Extracting dominant terms from the numerator and denominator,

$$\lim_{x \rightarrow \infty} \frac{x^8 + \ln x}{3x^8 + 2} = \lim_{x \rightarrow \infty} \frac{x^8}{3x^8} = \frac{1}{3}.$$

$$(m) \lim_{x \rightarrow \infty} \frac{x^{10} + 2x^2 + 10}{32x^6 + 12} =$$

Solution:

Extracting dominant terms from numerator and denominator,

$$\lim_{x \rightarrow \infty} \frac{x^{10} + 2x^2 + 10}{32x^6 + 12} = \lim_{x \rightarrow \infty} \frac{x^{10}}{32x^6} = \frac{1}{32} \lim_{x \rightarrow \infty} x^4 = \infty$$

$$(n) \lim_{n \rightarrow \infty} \frac{n^5 - 10n^7 + \ln n}{n^3 + 8n^7} =$$

Solution:

Extracting dominant terms from both the numerator and denominator,

$$\lim_{n \rightarrow \infty} \frac{n^5 - 10n^7 + \ln n}{n^3 + 8n^7} = \lim_{n \rightarrow \infty} \frac{-10n^7}{8n^7} = -\frac{5}{4}$$

(o) $\lim_{x \rightarrow \infty} \frac{e^x}{x^3} =$

Solution: As $x \rightarrow \infty$, the numerator grows faster than denominator, $\lim_{x \rightarrow \infty} \frac{e^x}{x^3} = \infty$

(p) $\lim_{x \rightarrow -\infty} (-3x^3 - 4x + 5) =$

Solution:

Since the leading term is $-3x^3$,
 $\lim_{x \rightarrow -\infty} (-3x^3 - 4x + 5) = \lim_{x \rightarrow -\infty} (-3x^3) = \infty$

(q) $\lim_{n \rightarrow \infty} \frac{n^{15} - 10n^7 + \ln n}{n^3 + 8n^7} =$

Solution:

Extracting leading terms from numerator and denominator,
 $\lim_{n \rightarrow \infty} \frac{n^{15} - 10n^7 + \ln n}{n^3 + 8n^7} = \lim_{n \rightarrow \infty} \frac{n^{15}}{8n^7} = \infty$

2. Show all the work to find limits for some common functions:

(a) $\lim_{x \rightarrow -\infty} e^x =$

Solution:

$$\lim_{x \rightarrow -\infty} e^x = \lim_{x \rightarrow \infty} \frac{1}{e^x} = 0$$

(b) $\lim_{x \rightarrow -\infty} x \ln x =$

Solution:

Since $-\infty$ is not in the domain of $x \ln(x)$, the limit does not exist.

(c) $\lim_{x \rightarrow \infty} x \ln x =$

Solution:

$$\lim_{x \rightarrow \infty} x \ln x = \lim_{x \rightarrow \infty} x \cdot \lim_{x \rightarrow \infty} \ln x = \infty \cdot \lim_{x \rightarrow \infty} \ln x = \infty$$

(d) $\lim_{x \rightarrow -\infty} e^{-x} =$

Solution:

$$\lim_{x \rightarrow -\infty} (-x) = \infty \implies \lim_{x \rightarrow -\infty} e^{-x} = e^{\lim_{x \rightarrow -\infty} (-x)} = \infty$$

(e) $\lim_{x \rightarrow \infty} e^{-x} =$

Solution:

$$\lim_{x \rightarrow \infty} e^{-x} = \lim_{x \rightarrow \infty} \frac{1}{e^x} = \frac{1}{\lim_{x \rightarrow \infty} e^x} = 0$$

(f) $\lim_{x \rightarrow \infty} \frac{1}{x} \sin x =$

Solution:

$$-1 \leq \sin x \leq 1 \implies -\frac{1}{x} \leq \frac{1}{x} \sin x \leq \frac{1}{x}.$$

$$\lim_{x \rightarrow \infty} -\frac{1}{x} = \lim_{x \rightarrow \infty} \frac{1}{x} = 0 \implies \lim_{x \rightarrow \infty} \frac{1}{x} \sin x = 0 \text{ by the Squeeze Theorem.}$$

(g) $\lim_{x \rightarrow -\infty} e^{\frac{1}{x}} =$

Solution:

$$\lim_{x \rightarrow -\infty} e^{\frac{1}{x}} = e^{\lim_{x \rightarrow -\infty} \frac{1}{x}} = e^0 = 1$$

(h) $\lim_{x \rightarrow \infty} e^{\frac{1}{x}} =$

Solution:

$$\lim_{x \rightarrow \infty} e^{\frac{1}{x}} = e^{\lim_{x \rightarrow \infty} \frac{1}{x}} = e^0 = 1$$

(i) $\lim_{x \rightarrow \infty} \frac{1}{e^{-x}} =$

Solution: $\lim_{x \rightarrow \infty} \frac{1}{e^{-x}} = \lim_{x \rightarrow \infty} e^x = \infty$

(j) $\lim_{x \rightarrow \infty} \ln x =$

Solution: $\ln x$ increases without bound as x approaches ∞ , $\lim_{x \rightarrow \infty} \ln x = \infty$

(k) $\lim_{x \rightarrow \infty} \sin x =$

Solution: $\sin x$, oscillates between -1 and 1, as $x \rightarrow \infty$, the limit doesn't exist.

(l) $\lim_{x \rightarrow \infty} \sin\left(\frac{1}{x}\right) =$

Solution: $\lim_{x \rightarrow \infty} \sin\left(\frac{1}{x}\right) = \sin\left(\lim_{x \rightarrow \infty} \frac{1}{x}\right) = \sin(0) = 0.$